



LOMBA KOMPETENSI SISWA
SEKOLAH MENENGAH KEJURUAN



TINGKAT KABUPATEN BEKASI

BEKASI, MEI 2024

SOAL
MODUL A – LINUX ENVIRONMENT
(TERBUKA)

BIDANG LOMBA
IT NETWORK SYSTEM ADMINISTRATOR
(IT NETWORK)



PEMERINTAH PROPINSI JAWA BARAT

DINAS PENDIDIKAN

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I. GENERAL INSTRUCTIONS

1. Carefully check the practice exam question documents, which consist of 3 page
2. Check the equipment and materials needed
3. Use the main equipment and work safety equipment that has been provided provided
4. Use equipment according to SOP (Standard Operating Procedure)
5. Work by paying attention to the instructions of the Judge

II. ASSIGNMENT PRACTICE QUESTIONS

WORK TASK INSTALLATION (BEKASISRV, CIKARANGSRV)

Note Please use the default configuration if you are not given details.

WORK TASK SERVER BEKASISRV

Configure the server with the hostname, domain and IP specified in the appendix.

o Configure the disk and partitions

- Add 3 disk with 7 GB of each disk.
- Use the three virtual disks to create a software RAID 5.
- Mount it as /files

o Create 30 local UNIX users with password “InaSkills 2019”

- Username: user[1-30]. ex: user1, user2, ..., user30

o Install the services:

1. DNS (bind9)

- Configure and install DNS Server with two domain
 - skills4future.net to BEKASISRV
 - skills39.edu.id to CIKARANGSRV
- Create subdomain files.skills4future.net and internal.skills4future.net
- Create subdomain monitor.skills39.edu.id and vpn.skills39.edu.id
- Create a host www.skills4future.net for IP Public JABARRO

2. Web Server (apache2 including php5)

- Create website “http://internal.skills4future.net” and “http://www.skills4future.net”
 - Use the following code for index.html in the http://internal.skills4future.net


```
<html>
      <h1>Welcome in the INTERNAL skills4future.net</h1>
</html>
```
 - Use the following code for index.html in the http://www.skills4future.net


```
<html>
      <h1>Welcome in the skills4future.net</h1>
</html>
```
- Make sure “http://internal.skills4future.net” is protected by authentication
 - Allow users from “user11” to “user20”
- Enable HTTPs for both sites
 - Use a certificate signed by CA Service in CIKARANGSRV
 - Make sure no certificate warning is shown.

3. FTP (proftpd)

- Enable FTPS
 - Use a certificate signed by CIKARANGSRV
- Each user (user21 to user30) will have a home directory
- Make sure the user is jailed in their respective website document root directories.
- Make sure file transfer to the server is possible.

4. Mail

- Make sure user11 to user20 have access via POP3, IMAP and SMTP
- Before you finish your project make sure you send an email message from user14 to user19 and another message from user19 to user14.
- Do not delete these email messages.

5. File Server (Samba)

- Share “MANAGER”
 - Path is /files/manager
 - Give access only to users “user1” to “user10”
 - Make sure the share is not shown in the network browser of the clients
- Share “GUEST”
 - Path is /files/guest
 - Enable read-only access to everyone

6. SSH Server

- Install SSH Server
- Use RADIUS CIKARANGSRV to authentication users.
- Change SSH port default to 1945

7. DHCP

- Create DHCP Pool INTERNAL:
 - Range: 192.168.150.51 – 192.168.150.100
 - Netmask: /25
 - Gateway: 192.168.150.1
 - DNS: 172.23.199.3
 - DNS-Suffix: skills4future.net
- SOLOCLT should always receive the following IP: 192.168.150.88
- The clients should automatically register their name with the DNS server after they
- have been assigned with an IP address by the DHCP server.